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Signals of large scale climate drivers, hatchery enhancement, and marine factors in Yukon River Chinook salmon survival revealed with a Bayesian life history model

Curry J. Cunningham  | Peter A. H. Westley | Milo D. Adkison

Department of Fisheries, College of Fisheries and Ocean Sciences, University of Alaska Fairbanks, Fairbanks, Alaska

Correspondence

Curry J. Cunningham, Department of Fisheries, College of Fisheries and Ocean Sciences, University of Alaska Fairbanks, Fairbanks, AK.
Email: cunninghamcurry@gmail.com

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Pollock Conservation Cooperative Research Center at the University of Alaska, Fairbanks

Abstract

Understanding how species might respond to climate change involves disentangling the influence of co-occurring environmental factors on population dynamics, and is especially problematic for migratory species like Pacific salmon that move between ecosystems. To date, debate surrounding the causes of recent declines in Yukon River Chinook salmon (*Oncorhynchus tshawytscha*) abundance has centered on whether factors in freshwater or marine environments control variation in survival, and how these populations at the northern extremity of the species range will respond to climate change. To estimate the effect of factors in marine and freshwater environments on Chinook salmon survival, we constructed a stage-structured assessment model that incorporates the best available data, estimates incidental marine bycatch mortality in trawl fisheries, and uses Bayesian model selection methods to quantify support for alternative hypotheses. Models fitted to two index populations of Yukon River Chinook salmon indicate that processes in the nearshore and marine environments are the most important determinants of survival. Specifically, survival declines when ice leaves the Yukon River later in the spring, increases with wintertime temperature in the Bering Sea, and declines with the abundance of globally enhanced salmon species consistent with competition at sea. In addition, we found support for density-dependent survival limitations in freshwater but not marine portions of the life cycle, increasing average survival with ocean age, and age-specific selectivity of bycatch mortality in the Bering Sea. This study underscores the utility of flexible estimation models capable of fitting multiple data types and evaluating mortality from both natural and anthropogenic sources in multiple habitats. Overall, these analyses suggest that mortality at sea is the primary driver of population dynamics, yet under warming climate Chinook salmon populations at the northern extent of the species' range may be expected to fare better than southern populations, but are influenced by foreign salmon production.

KEYWORDS

Bayesian estimation, Chinook salmon conservation, climate change, environmental regulation of survival, life cycle models, marine and freshwater mortality, salmon assessment, Yukon River

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1 | INTRODUCTION

Predicting species' responses to ongoing and future climate change hinges on understanding the role of past environmental variation (Stenseth et al., 2002). However, it is challenging to disentangle the influence of multiple interacting factors on population processes that occur at varying scales in time and space (Levin, 1992). Compounding this challenge is the need to identify critical periods in the life cycle where inter- and intra-specific density-dependent capacity limitations result in survival bottlenecks, and thereby govern population dynamics. To robustly identify the relationship between local- and regional-scale climate and physical processes and observed variation in survival rates, it is necessary to develop statistical models capable of testing alternative hypotheses while also accounting for anthropogenic impacts such as direct or incidental harvest of exploited species. Quantifying environmental regulation of survival is even more challenging for species that cross ecosystem boundaries and migrate between distinct habitats (Piou & Prevost, 2013).

One example is salmon, whose anadromous life history necessitates the evaluation of both freshwater and marine processes on survival, and where harvest in directed fisheries and capture as non-target bycatch requires a holistic approach to quantifying both natural and anthropogenic sources of mortality. Pacific salmon play a unique and important role in ecosystems of the North Pacific. Their migrations move energy and nutrients between marine and freshwater or terrestrial environments (Helfield & Naiman, 2001; Hocking & Reynolds, 2012; Holtgrieve, Schindler, & Jewett, 2009), providing seasonal subsidies for a wide range of species (Bilby, 1996; Field & Reynolds, 2011; Hilderbrand et al., 1999; Scheuerell, Moore, Schindler, & Harvey, 2007). Chinook salmon (*Oncorhynchus tshawytscha*) populations returning to the Yukon River of Alaska provide vast cultural and economic value to indigenous people of the region that subsist on the resource (Brown & Godduhn, 2015). Although northern latitude ecosystems like the Yukon River basin are experiencing some of the most rapid climate change on the globe (IPCC, 2014), we know little about how Chinook salmon populations in Alaska are likely to respond to climate change relative to populations in the southern portion of the species' range (Michel et al., 2015; Ward, Anderson, Beechie, Pess, & Ford, 2015; Yates et al., 2008; Zabel, Scheuerell, McClure, & Williams, 2006; Zeug, Albertson, Lenihan, Hardy, & Cardinale, 2011). Flowing more than 3,000 km from Canada to the Bering Sea (Eiler, Masuda, Spencer, Driscoll, & Schreck, 2014), the Yukon River is the fourth largest drainage in North America (855,000 km²), and until recently was the largest producer of Chinook salmon in the region (Carey, 1980; Gilbert & O'malley, 1921). During the period 1961–2014, the annual harvest averaged 128,740 Chinook salmon and total run size (catch plus those fish that had the opportunity to spawn) averaged near 300,000 fish between 1982 and 1997 (JTC, 2016, Schindler et al., 2013). However, Chinook salmon returns to the Yukon watershed declined by 45% between 1998 and 2010 relative to the preceding 1982–1997 period (Schindler et al., 2013).

This recent period of low Yukon Chinook salmon production has resulted in severe limitations to commercial and subsistence fisheries in the region and difficulty in meeting international treaty obligations for Chinook salmon escapement to Canada, and prompted their listing as a stock of yield concern by the Alaska Board of Fisheries in 2000. The total harvest of Yukon Chinook salmon by both United States and Canadian users declined by 57% after 1997 (JTC, 2016). Among user groups, the US commercial harvest of Yukon Chinook salmon has experienced the largest decline (74%) during the post-1997 period, with complete closure of Alaskan commercial fisheries for Yukon Chinook salmon in 2012 (JTC, 2016). Difficulty meeting the minimum number of spawning individuals required for portions of the Yukon River since 2008 has also resulted in severe restriction of Alaska subsistence harvest, to below the level deemed necessary by the Alaska Board of Fisheries for rural native communities, threatening both food security and a way of life maintained for millennia (Brown & Godduhn, 2015; Schindler et al., 2013). These reductions in Chinook salmon subsistence fishing opportunities combined with low return abundance of chum salmon (*O. keta*), has necessitated the designation of the Yukon River as either a fishery disaster or economic disaster nine times since 1997 by NOAA Fisheries.

Given the importance of subsistence and commercial fisheries for Yukon River Chinook salmon, and the difficulty in managing these fisheries equitably in a trans-boundary river, there is intense interest in understanding the factors driving variation in Chinook salmon survival in both freshwater and marine environments and how populations are likely to respond to future climate change. To date, most studies have examined the effect of freshwater or marine processes in isolation, although salmon span ecosystem boundaries. Neuswanger, Wipfli, Evenson, Hughes, and Rosenberger (2015) evaluated whether a range of habitat variables in the *freshwater* environment could explain observed recruitment patterns for two index stocks of the Middle Yukon, finding that lower productivity of these populations was associated with increased water discharge in natal rivers during juvenile rearing, and postulated that this effect resulted from reduced foraging efficiency during high flow events. Taking a broader perspective, Ohlberger, Scheuerell, and Schindler (2016) investigated coherence in Chinook salmon population dynamics across space, using Dynamic Factor Analysis methods to identify shared trends in the productivity of 15 Alaskan Chinook salmon populations across time and to estimate the influence of indicators in both freshwater and marine environments. With respect to the productivity of the Middle Yukon stocks (the focus of our paper), Ohlberger et al. (2016) found a negative effect of ice out date, a negative but nonsignificant effect of Russian Chinook salmon catch, and a positive but nonsignificant effect of the North Pacific Gyre Oscillation (NPGO).

Evaluation of marine mortality sources for Yukon Chinook salmon has primarily focused on the impact of bycatch in other Bering Sea and Aleutian Islands fisheries (Stram & Harrington, 2009). To quantify the impact of bycatch in the Bering Sea pollock (*Gadus chalcogrammus*) trawl fishery on western Alaska Chinook salmon production, Ianelli and Stram (2014) created an adult equivalency (AEQ)

model to estimate region-specific impact rates, or proportional losses in Chinook salmon returns after accounting for subsequent age- and year-specific natural mortality. Bycatch mortality was estimated to have averaged 853 (range: 300–1,717) Chinook salmon per year for the Middle Yukon between 1994 and 2012, peaking in 2007. AEQ bycatch mortality for the Upper Yukon during this period averaged 1,771 (610–3,024), representing an annual impact rate (i.e., percent of foregone returns) of 1.74% (0.79–4.03%) for this group of populations (Ianelli & Stram, 2014). Thus, while bycatch does reduce the total number of potentially surviving fish, it does not appear to be the primary driver of declining salmon returns to the Yukon River basin. However, the impact of bycatch mortality in the context of environmentally driven survival variation has not been evaluated.

While successful in testing specific hypotheses about the drivers of variation in Yukon River Chinook salmon production, these previous analyses either: (a) only evaluated hypotheses pertaining to the freshwater environment (Neuswanger et al., 2015), (b) evaluated the influence of anthropogenic (Ianelli & Stram, 2014) or natural (Neuswanger et al., 2015) factors in isolation, (c) did not incorporate potential data sources including juvenile indices of abundance and bycatch numbers and age composition to inform estimation (Ohlberger et al., 2016), or (d) did not account for the influence of observed changes in female age composition and sex ratio over time (JTC, 2016; Schindler et al., 2013) on spawning potential.

In light of these previous analyses, our goal was to evaluate competing hypotheses about the natural and anthropogenic drivers of variation in Yukon Chinook salmon survival in *both* freshwater and marine environments and assess the strength and location of density-dependent capacity limitations, utilizing all available sources of juvenile and adult abundance, age composition, and bycatch data to inform estimation. To accomplish this goal we constructed a novel Bayesian stage-structured population dynamics model, which used Bayesian model selection methods to quantify the level of support from the data for each candidate covariate (O'hara & Sillanpaa, 2009) and Bayesian model averaging (Hoeting, Madigan, Raftery, & Volinsky, 1999; Raftery, Madigan, & Hoeting, 1997) to account for model structural uncertainty in predictions. Finally, we quantified the influence of climate and other factors by simulating the expected differences in survival resulting from fixed changes in the value of each environmental factor, while accounting for both estimation and model structural uncertainty.

2 | MATERIALS AND METHODS

2.1 | Population dynamics model

The population dynamics model tracked cohorts of Chinook salmon from specific brood years forward in time across sequential life stages. The number of Chinook salmon surviving through the current stage (s) depended upon the abundance of individuals from brood year y , surviving the previous stage $s-1$, of population p ($N_{y,s-1,p}$), and the year, stage, and population-specific survival rate $SR_{y,s,p}$:

$$N_{y,s,p} = N_{y,s-1,p} SR_{y,s,p} \quad (1)$$

Survival rate through each life stage (Figure 1) was represented by a Beverton-Holt transition function (Moussalli & Hilborn, 1986).

The Beverton-Holt equation (Beverton & Holt, 1957) provides a useful approximation for density-dependent survival of individuals, and has been utilized in a wide range of life cycle models for Chinook salmon (see Scheuerell et al., 2006; Honea et al., 2009, and Battin et al., 2007). The survival rate was calculated as:

$$SR_{y,s,p} = \frac{p_{y,s,p}}{1 + \frac{p_{y,s,p} N_{y,s-1,p}}{k_{s,p}}} \quad (2)$$

where the productivity parameter ($p_{y,s,p}$) represented the maximum survival rate in the absence of density-dependent compensation, and the capacity parameter ($k_{s,p}$) described the maximum number of individuals that could potentially survive through the stage.

Stage and population-specific productivity parameters ($p_{y,s,p}$) were treated as time-varying functions of the environmental covariates under evaluation. Each productivity parameter was an inverse-logit function of a basal productivity $\beta_{s,p,0}$, or the average survival rate at low density across years (in logit space), and the sum of covariate effects

$$p_{y,s,p} = \frac{1}{1 + \exp\left(-\beta_{s,p,0} - \sum_{c=1}^{Nc_{s,p}} \beta_{p,c} X_{t,c}\right)} \quad (3)$$

$$t = y + \delta_c$$

where covariate c values at time t ($X_{t,c}$) were multiplied by coefficients ($\beta_{p,c}$) describing their respective influence on stage and population-specific survival rates. The parameter δ_c defined the difference between the brood year y and the calendar year t in which the cohort interacted with each covariate, and $Nc_{s,p}$ was the number of productivity covariates linked to each population in each stage (See Table 1 for a description of covariates and stage associations, Supporting Information Figure S1 for the time series of covariate values).

The Chena and Salcha Chinook salmon populations of the Middle Yukon River were selected as index populations given the breadth of data available and their use in previous evaluations of Yukon River Chinook salmon production dynamics (see Neuswanger et al., 2015 and Ohlberger et al., 2016). These populations are among the largest contributors to the total US Chinook salmon production in the Yukon River and have been extensively monitored for decades (Brown, Finster, Henszey, & Eiler, 2017). Individuals from these populations primarily spend 1 year in freshwater before migrating to sea in the spring and summer (brood year + 2) and remain at sea for 1–5 years before maturing, primarily returning as 5 and 6-year olds (European Age: 1.3 and 1.4).

The life cycle of these populations was modeled as seven stages describing distinct locations, time periods, or ontogenetic phases: (1) “upstream” tracking survival during juvenile rearing in tributaries in brood year +1, (2) “downstream/nearshore” describing survival

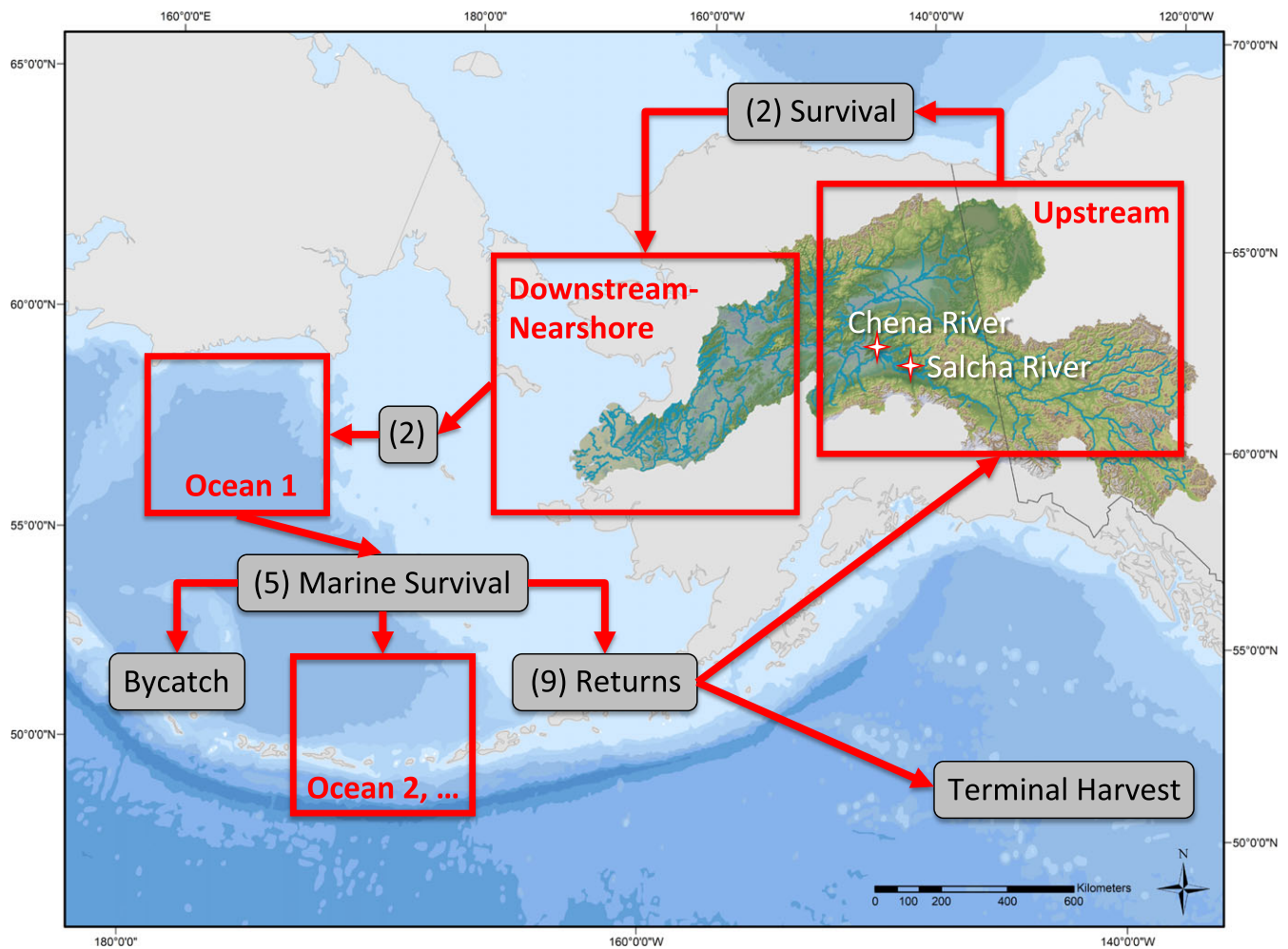


FIGURE 1 Schematic map of the Yukon River Chinook salmon population dynamics model. Red boxes identify model stages and gray boxes describe stage transitions or removals from the system. Numbers in parenthesis reference model equations described in the text. It should be noted that red boxes describing model stages are meant to generally reflect model structure and spatial orientation, but do not reference the exact location of Chinook salmon at any given time

through the Yukon mainstem and early ocean entry (brood year +2) prior to the juvenile offshore survey, and (3–7) separate stages representing survival through five potential years at sea (Figure 1). Two capacity parameters ($k_{s,p}$) were estimated for each population, one for the freshwater (1–2) and the other for the marine (3–7) model stages, based on the a priori expectation that capacity limits should be different in these two environments. Conversely, separate basal productivities ($\beta_{s,p,0}$) were estimated for each freshwater stage (1 and 2) and the first marine stage (3), while the basal productivities for subsequent marine stages (4–7) were a linear function of the basal productivity in the first marine stage (3),

$$\beta_{s \in 4:7,p,0} = \beta_{s=3,p,0} + \omega_p(s-3) \quad (4)$$

where ω_p is the change in basal productivity across marine stages. The formulation of marine basal productivities (Equation 4) allowed for a monotonic change in marine survival as Chinook aged, while reducing the number of estimated parameters by three. In addition, this permitted us to test the hypothesis that the base survival rate

for Chinook salmon changed as a function of growth at sea, independent of time-varying environmental effects.

The abundance ($N_{y,s,p}$) of Chinook salmon cohorts was tracked forward in time across model stages based on predicted survival rates (Equations 2 and 3), independent of whether the stage was in the freshwater or marine portion of the life cycle. However, for marine model stages it was also necessary to account for both the maturation process and bycatch mortality in the BSAI pollock fishery (Ianelli & Stram, 2014; Stram & Ianelli, 2014). Given that natural and bycatch mortality are operating concurrently, this was represented as a continuous process (Doubleday, 1976; Pope, 1974). The number of individuals surviving through an ocean stage:

$$N_{y,s,p} = N_{y,s-1,p} e^{-(S_{a,p} F_{t,p} + M_{y,s,p})} (1 - \phi_{a,p}) \quad (5)$$

$$M_{y,s,p} = -\ln(\text{SR}_{y,s,p})$$

was a function of time-varying natural mortality ($M_{y,s,p}$) and the instantaneous fishing mortality rate ($F_{t,p}$) in calendar year t for population p , scaled by the ocean-age a

TABLE 1 Description of environmental covariates. Months are the period for which each covariate is derived, while the brood year offset describes the difference between the year of covariate influence and the brood year for each cohort. Covariate values were standardized based on the mean and standard deviation (SD) of each time series

Covariate	Description	Months	Data Origin	Website	Stage	Brood Year Offset	Mean	SD
chena.discharge	Average monthly discharge for Chena River (cfs ⁻¹)	May - August	USGS Guage #15493000: CHENA R NR TWO RIVERS AK	http://waterdata.usgs.gov/nwis/inventory?agency_code=USGS&site_no=15493000	Upstream	1	1354	415
salcha.discharge	Average monthly discharge for Salcha River (cfs ⁻¹)	May - August	USGS Guage #15484000: SALCHA R NR SALCHAKET AK	http://waterdata.usgs.gov/nwis/inventory?agency_code=USGS&site_no=15484000		1	3062	824
ice.tanana	Date of ice out on Tanana River at Nenana (DOY)		Nenana Ice Classic: Tanana River Ice Annual Breakup Dates, Version 1. 1960-2015. Boulder, Colorado USA. NSIDC	http://nsidc.org/data/NSIDC-0064/versions/		1	121.8	5.9
temp.incub	Average air temperature during incubation (°F)	November - February	Western Regional Climate Center	http://www.wrcc.dri.edu/cgi-bin/cliMAIN.pl?ak2968		0	-2.43	3.91
temp.juvi	Average air temperature during juvenile rearing (°F)	May - August				1	57.43	1.78
ice.yukon	Date of ice out on Yukon River at Dawson, YT (DOY)			http://www.yukonriverbreakup.com/statistics	Downstream/ Nearshore	2	125.2	4.7
wind.stress.spring	Along-peninsula component of the wind stress in spring (N m ⁻²)	May - June	Atmosphere Indices, Bering Climate, at_UPWs.nc Stabeno et al. (2002)	http://www.beringclimate.noaa.gov/data/index.php		2	0.03	0.07
wind.mixing.june	Mean wind mixing index at M2 [*]	June - July	Atmosphere Indices, Bering Climate, at_WndMix.nc Stabeno et al. (2001)			2	0.0152	0.0048
se.bs.may.sst	Mean May sea surface temperature in the southeastern Bering Sea	May	Ocean Indices, Bering Climate, oc_MaySST.nc			2	2.18	0.85
arctic.dipole.winter	Winter Arctic Dipole index	December - February	Climate Indices, Bering Climate, in_ADWin.nc			2	0.02	0.54
slp.spring	Mean spring sea level pressure in the Bering Sea	April - June	Atmosphere Indices, Bering Climate, at_BSPIsP.nc			2	1101	2.1
ice.retreat	Ice Retreat Index: Number of days with ice cover after March 15 at the M2 [*]	March - May	Ocean Indices, Bering Climate, oc_IRI.nc Hunt et al. (2011)	Ocean 1	3	17.29	17.51	
winter.prib.temp	Sea surface temperature at the Pribilof Islands (°C)	January - March	Ocean Indices, Bering Climate, oc_PISST.nc		3	-0.058	0.29	
m2.fav.wind	Number of days with winds between 4.8 and 9.5 m/s from at M2 [*]	May - mid-July	Atmosphere Indices, Bering Climate, at_NODM2.nc Megrey and Hinckley (2001)		3	37.06	4.49	
slp.winter	Mean winter sea level pressure in the Bering Sea	December - March	Atmosphere Indices, Bering Climate, at_BSPIsW.nc		3	1006.27	2.01	
npgo	Annual average North Pacific Gyre Oscillation index		Di Lorenzo et al (2008)		3	0.11	1.06	
wind.stress.winter	Along-peninsula component of the wind stress in winter (N m ⁻²)	November - April	Atmosphere Indices, Bering Climate, at_UPW.nc		3	0.01	0.05	
arctic.oscillation	Winter Arctic Oscillation Index	December - February	Climate Indices, Bering Climate, in_AO.nc		3	0.18	1.02	
el.nino	Winter multivariate ENSO index	December - January	Climate Indices, Bering Climate, in_ENSO.nc		3	0.19	0.66	
alaskan.index	Alaska Index, relative to 1961-2000 index values	December - March	Climate Indices, Bering Climate, in_Al.nc		3	0.17	0.86	
jpn.hat.chum	Japanese hatchery chum abundance (millions)		Ruggerone and Irvine (2015), North Pacific Anadromous Fish Comission		http://www.npafc.org/new/publications/Documents/PDF%202015/1594(USA+Canada).pdf	3	56.44	14.57
pink.total	Total North Pacific pink salmon abundance (wild + hatchery, in millions)				3	388.08	103.43	

specific selectivity ($S_{a,p}$). The calendar year t was the brood year plus an offset for the specific stage ($t = y + p_s$), the ocean age was the current stage s minus the number of freshwater stages ($a = s - n_{fw}$), and ϕ_a was the ocean-age specific probability of maturing. The age-specific maturation probability was represented by a modified logistic function:

$$\phi_{a,p} = \frac{1}{1 + e^{\lambda_p(5-a) + \ln(1/0.99-1)}} \quad (6)$$

where λ_p was a population-specific parameter controlling the maturation schedule, and estimated from the age composition data and assumed to be time-invariant. This simplified logistic function ensures the majority of remaining individuals mature in their fifth year at sea so as to agree with the maximum marine residency observed in these populations and simulated in the model, and requires estimation of one less parameter than the standard logistic function.

Immature and maturing Yukon River Chinook salmon are caught in Bering Sea and Aleutian Islands groundfish trawl fisheries. Bycatch of Chinook salmon in the BSAI pollock trawl fishery was calculated as,

$$C_{t=y+p_s, a=s-n_{fw}, p} = C_{y,s,p} = \left(\frac{S_{a,p} F_{t,p}}{M_{y,s,p} + S_{a,p} F_{t,p}} \right) N_{y,s-1,p} \left(1 - e^{-(S_{a,p} F_{t,p} + M_{y,s,p})} \right) \quad (7)$$

and described by brood y or calendar t year, and stage s or ocean age a , with appropriate offsets. Chinook salmon bycatch data from BSAI trawl fisheries were not available prior to 1991 to inform estimation of bycatch fishing mortality rates ($F_{t,p}$); therefore, annual fishing mortality rates were treated as random effects drawn from a population-specific lognormal distribution.

$$F_{t,p} \sim \text{log normal}(\mu_p, v_p) \quad (8)$$

Hyperparameters μ_p and v_p are the lognormal mean and variance describing the expected distribution of annual bycatch fishing mortality rates, effectively specifying the bycatch rate in the absence of observation, and were estimated from the data.

The number of individuals returning to spawn, termed “spawn-ers”, ($R_{t,a,p}$) was calculated as the number surviving marine and bycatch mortality who then matured, less those caught in terminal commercial, recreational, and subsistence fisheries,

$$R_{t=y+p_s, a=s-n_{fw}, p} = N_{y,s-1,p} e^{-(S_{a,p} F_{t,p} + M_{y,s,p})} (\phi_{a,p}) (1 - U_{t,p}) \quad (9)$$

where $U_{t,p}$ is the year-specific terminal harvest rate estimated by Saveriede and Huang (2014) for the Chena and Salcha populations. $U_{t,p}$ was estimated outside of our model and assumed to be known without error. The number of returning individuals $R_{t,a,p}$ was the model-predicted spawning abundance.

The number of eggs produced for a given number of returning individuals was a function of the observed sex ratio, or the female proportion of the spawning abundance for each population in each year $\pi_{t,p}$, and average observed age-specific fecundity $f_{a,p}$.

$$E_{t,p} = \pi_{t,p} \sum_{a=1}^5 (R_{t,a,p} f_{a,p}) \quad (10)$$

Population-specific fecundity at age $f_{a,p}$ was found by regressing observed fecundity (Jasper & Evenson, 2006) against length for

each population and predicting fecundity for the average observed length of each ocean age class. The abundance of juveniles in the next cohort that survived to the end of the first stage was simply the total egg production ($E_{t,p}$) multiplied by the appropriate survival rate.

Both the terminal harvest rate and age-specific fecundity were estimated outside of our model and assumed known without error and treated as fixed constants.

The abundance of juveniles in the next cohort that survived to the end of the first stage was simply the total egg production ($E_{t,p}$) multiplied by the appropriate survival rate (Equation 11). Population dynamics were modeled for brood years 1980–2007 (return years: 1983–2014). Spawning abundances prior to the beginning of the time series of escapement observations (1980–1986) were needed to seed the initial population dynamics and were estimated as independent parameters based on their contribution to escapement in subsequent years.

$$N_{y=t,s=1,p} = S_{y,s=1,p} E_{t,p} \quad (11)$$

2.2 | ESTIMATION METHODS AND DATA

The population dynamics model was fit to available data using Bayesian methods, and implemented in JAGS (Plummer, 2013) using the R2jags (Su & Yajima, 2015) package in R. Models were fit to data for each population separately, with five chains run for 1,000,000 iterations with a thinning rate of 1/500. The initial 500,000 iterations were discarded as a burn-in period. Convergence of the chains was diagnosed using the Gelman-Rubin statistic (Brooks & Gelman, 1998) and visual inspection of the coherence in the distribution of posterior samples from each chain.

A primary goal of this analysis was to integrate all available sources of data in our assessment of Yukon Chinook salmon population dynamics. Therefore, estimation of model parameters was informed by fitting to five separate data types with appropriate likelihoods. Two indices of abundance were available for the Chena and Salcha populations: (i) spawning abundance ($A_{t,p}$) of population p in year t from mark-recapture experiments and counting tower observations (Saveriede & Huang, 2014), and (ii) absolute abundance of juvenile Chinook salmon from the Bering Aleutian Salmon International Survey ($J_{t,p}$), operated by the NOAA Alaska Fisheries Science Center since 2003 (Murphy et al., 2016). The juvenile index of abundance provided by Murphy et al. (2016) was not specific to the Chena and Salcha populations, so this index needed to be scaled based on available data. Genetic stock identification (GSI) data (J. Murphy, unpublished data) was used to translate the total juvenile catch per unit effort into an index of juvenile Chinook salmon from the Middle Yukon River (MYR). Population-specific indices of juvenile abundance were calculated from the MYR index based on the observation that an average of 42.7% of radio-tagged MYR Chinook salmon returned to the Tanana drainage (Eiler et al., 2014), and the average proportion of total Tanana escapement to the Chena (0.397)

and Salcha (0.603) rivers. Observation error for both adult and juvenile indices of abundance was assumed log-normally distributed,

$$\begin{aligned} \ln(A_{t,p}) &\sim \text{Normal}\left(\ln\left(\hat{A}_{t,p} = \sum_{a=1}^5 R_{t,a,p}\right), \sigma_{\text{adult},p}^2\right) \\ \ln(J_{t,p}) &\sim \text{Normal}\left(\ln\left(\hat{J}_{t,p} = N_{y=t-\rho_{s=2},s=2,p}\right), \sigma_{\text{juvenile},p}^2\right) \end{aligned} \quad (12)$$

with log-normal variance parameters for the juvenile ($\sigma_{\text{juvenile},p}^2$) and adult ($\sigma_{\text{adult},p}^2$) indices estimated from the data. The model was fit to spawning abundance data 1987–2010 and the juvenile index of abundance 2003–2009.

Chinook salmon bycatch data for the BSAI pollock fishery were available from the North Pacific Fisheries Management Council and total annual bycatch was allocated to MYR bycatch using GSI data (Guthrie, Nguyen, & Guyon, 2016; Ianelli & Stram, 2014), and scaled to the population level as described above. Bycatch observation error was assumed log-normally distributed, when fit to bycatch data 1991–2010,

$$\ln(c_{t,p}) \sim \text{Normal}\left(\ln\left(\sum_{a=1}^5 C_{t,a,p}\right), \sigma_{\text{bycatch},p}^2\right) \quad (13)$$

where $c_{t,p}$ is the observed bycatch in calendar year t scaled to the population level, and $\sigma_{\text{bycatch},p}^2$ is the population-specific log-normal variance parameter estimated from the data.

Age composition data were available from samples of both adult Chinook salmon on the spawning grounds and BSAI bycatch (Ianelli & Stram, 2014). Bycatch at age was calculated by Ianelli and Stram (2014) based on time-area stratified and combined age-length keys and bycatch length compositions from the fishery observer program. The estimation model was fit to age composition data for spawning adults 1987–2010 and bycatch 1991–2010 assuming the sampling process is best described by a Dirichlet-multinomial mixture,

$$\begin{aligned} \hat{P}_{i=\text{adult},t,a,p} &= R_{t,a,p} / \sum_{a=1}^5 R_{t,a,p} \\ \hat{P}_{i=\text{bycatch},t,a,p} &= C_{t,a,p} / \sum_{a=1}^5 C_{t,a,p} \\ \alpha_{i,t,a,p} &\sim \text{Dirichlet}(w_i \cdot \hat{P}_{i,t,a,p}) \\ S_{i,t,a,p} &\sim \text{Multinomial}\left(a_{i,t,a,p}, \sum_{a=1}^5 S_{i,t,a,p}\right) \end{aligned} \quad (14)$$

where $\hat{P}_{i,t,a,p}$ are the model-predicted age composition proportions from sample type i , $S_{i,t,a,p}$ are the spawning ground or bycatch compositions, and w_i is the weighting parameter for each sample type. The Dirichlet-multinomial likelihood has been advanced as a superior alternative to traditional multinomial likelihood for compositional data, because it allows for the overdispersion associated with the sampling process (Francis, 2014; Thorson, Johnson, Methot, & Taylor, 2016). Following the suggestions by Francis (2011), we chose to fix the weighting parameters at a level that reduced conflict between abundance and age composition data ($w_{i=\text{adult}} = 50, w_{i=\text{bycatch}} = 5$), rather than using model-based estimates of w_i that tended to preferentially fit age composition data. Optimal fixed values for Dirichlet-multinomial weights were found by iteratively fitting the model while increasing weights, until the point that the model's ability to fit the two abundance indices

was negatively impacted. While there is still extensive debate within the stock assessment literature about optimal methods for the weighting of composition data (see. Francis, 2017; Hulson, Hanselman, & Quinn, 2012; Punt, 2017), we believe the iterative method employed here provides a reasonable balance for inference from both age composition and abundance data.

2.3 | BAYESIAN MODEL SELECTION

To select the most parsimonious set of explanatory covariates (Table 1, Supporting Information Figure S1) and generate predictions that are robust to model structural uncertainty, we used Bayesian model selection methods. Specifically, we employed a random effects version of the indicator variable method described by Kuo and Mallick (1998), where the covariate influences in Equation (3) ($\beta_{p,c}$) are the product of the estimated covariate effect ($\epsilon_{p,c}$) and an indicator variable ($\tau_{p,c}$):

$$\begin{aligned} \beta_{p,c} &= \tau_{p,c} \epsilon_{p,c} \\ \tau_{p,c} &\sim \text{Bernoulli}(v_p) \end{aligned} \quad (15)$$

The indicator variable is Bernoulli distributed with a population-specific prior inclusion probability of v_p that was estimated as a hyperparameter (Table 2). For each sample from the posterior, $\tau_{p,c}$ takes a value of 0 or 1, functionally including or excluding that covariate in each step. As such, the average value of $\tau_{p,c}$ across posterior samples represents the inclusion probability for each covariate for each population, or the weight of evidence from the data for its inclusion. For this analysis, we implemented a random effects version of the Kuo and Mallick (1998) method by placing a hyperprior on γ_p , the standard deviation of the normal prior on covariate effects $\epsilon_{p,c}$ (Table 2). It has been suggested that the random effects prior structure is optimal because it helps in combatting Lindley's paradox (Lindley, 1957) and the resulting potential for underselection of candidate variables (Lawrence, Gramacy, Thomas, Buckland, & Freckleton, 2013; Link & Baker, 2006). In their review and comparison of Bayesian variable selection methods, O'hara and Sillanpaa (2009) found the random effects version of the Kuo and Mallick (1998) method to be unbiased and have lower variance in estimated effects, when compared to alternative Bayesian model selection methods. Most importantly, Raftery et al. (1997) highlight that ignoring model structural uncertainty results in underestimation of uncertainty in quantities of interest, and Bayesian model averaging across a set of models is likely to result in better inference than any single model in most cases (Hoeting et al., 1999; Madigan & Raftery, 1994).

2.4 | PRIORS

Prior distributions for estimated model and hyperparameters (Table 2) were chosen to be uninformative or weakly informative when transformations were considered. The exception was the hyperprior for the prior on covariate inclusion probability v_p , which

was Beta distributed to scale 0–1 with an expected value of 0.2 (95% probability range 0.03–0.48). By assuming this weakly informative prior on v_p , rather than using a fixed value, we allowed the model to learn about the optimal degree of sparsity in terms of covariate inclusion while providing a moderate penalty on overfitting. The prior specified for ω_p , the change in basal productivity across ocean stages (Equation 4), was normally distributed around zero indicating an a priori hypothesis of no change in survival across marine years.

2.5 | COVARIATES

The environmental covariates included in this analysis (Table 1, Supporting Information Figure S1) represent a priori hypotheses for the processes driving variation in Yukon River Chinook salmon survival rates. We evaluated five covariates describing four hypotheses about survival during incubation and early juvenile rearing, each relating to habitat conditions and the onset of primary production in spring, as approximated by the date of ice breakup. Separate water discharge data were available for the Chena and Salcha rivers, while all other covariates were not stock specific. Covariates for the downstream and nearshore stage addressed conditions in the lower Yukon River and the Bering Sea at ocean entry, prior to the juvenile offshore survey (Table 1, Supporting Information Figure S1). These included the date of Yukon River ice out in the year of ocean entry, the Arctic Dipole Index during the months preceding ocean entry, and the Bering Sea sea level pressure (SLP) and sea surface temperature (SST) in the spring of ocean entry. Also explored were springtime wind stress, which quantifies transport of nutrient-rich waters through Unimak Pass (Stabeno, Reed, & Napp, 2002), and the level of wind mixing in June, which plays an important role in determining whether nutrient-rich water from lower in the water column will be moved into the photic zone and foster continued plankton blooms during the summer (Stabeno, Bond, Kachel, Salo, & Schumacher, 2001).

Marine environmental covariates quantified a wide range of oceanographic, atmospheric, climatic, and biotic factors (Table 1, Supporting Information Figure S1). Oceanographic processes included the timing of ice retreat in the southeastern Bering Sea, given implications for the timing and biomass of the spring plankton bloom and trophic structure of the Bering Sea (Hunt et al., 2002, 2011), and winter SST at the Pribilof Islands given its observed influence on western Alaska Chinook salmon growth (Mcphee, Leon, Wilson, Siegel, & Agler, 2016; Siegel, Mcphee, & Adkison, 2017) and correlation with recruitment of several groundfish species (Mcfarlane, King, & Beamish, 2000) as potential prey. Covariates related to the marine atmosphere included SLP and wind stress during the first winter at sea given their observed importance for ecosystem productivity, and the duration of May–early June wind vectors in the southern Bering Sea (at the NOAA-PMEL M2 OA mooring: 56.87°N, –164.06°W) that promote favorable feeding conditions for larval pollock (Megrey & Hinckley, 2001). Marine climate-related covariates included the North Pacific

Gyre Oscillation (Di Lorenzo et al., 2008), which is correlated with fluxes in salinity, nutrients and *chlorophyll-a* concentrations, the Arctic Oscillation Index describing dominant trends in North Pacific SLP, and the Alaska Index which measures the atmospheric circulation driving variation in ice cover.

Two covariates were included to test the hypothesis that competition with other salmonid species in the marine environment may influence survival: the return abundance of Japanese hatchery chum salmon (*Oncorhynchus keta*) and the total North Pacific return abundance of pink salmon (*Oncorhynchus gorbuscha*; hatchery and wild). Return abundances were included rather than hatchery releases because we believe the return abundance better quantifies potential competition, compared with hatchery releases which are separated from the period of potential interaction in the Bering Sea by the period of highly variable early marine survival.

All covariates were z-standardized by subtracting the mean and dividing by the standard deviation of the 1980–2010 time series. The covariates included in these analyses were prescreened to ensure that absolute Pearson correlation among covariates did not exceed 0.6 (Table 1, Supporting Information Figure S1).

2.6 | SENSITIVITY ANALYSIS: SURVIVAL RATES AND POPULATION PRODUCTIVITY

We conducted a sensitivity analysis to compare the effects of different covariates on expected survival and population productivity (recruits-per-spawner), while accounting for model structural uncertainty (e.g., inclusion probabilities), estimation uncertainty, bycatch mortality, and survival through subsequent life stages. We simulated Bayesian model averaged predictions by: (a) Drawing 5,000 sets of model parameters from the joint posterior to preserve posterior correlation (including indicator variables controlling parameter inclusion), (b) Simulating survival through one complete life cycle for each of these parameter sets while iteratively increasing each covariate by one standard deviation from its long-term mean, and (c) Comparing results to a “base case” with all covariates at their long-term (31-year) mean. In all scenarios the average spawning abundance, proportion of females, age of females, and adult harvest rate were used, as well as the predicted mean for the distribution of hierarchical bycatch fishing mortality rates.

3 | RESULTS

Estimated effects and inclusion probabilities for covariates together suggest that variation in the survival of the Chena and Salcha populations is better explained by processes in the marine rather than the freshwater environment. A comparison of model predictions to the observed data for both populations (Figure 2) shows that the model generally captured the variation in all data sources with reasonable accuracy and limited bias. Of note is a tendency for the model to underpredict the proportion of 2-ocean individuals in the escapement of both populations, and to a lesser extent

TABLE 2 Estimation model parameter descriptions and specified priors

Parameter	Description	Prior
$\beta_{s \in 1:3,p,0}$	Basal productivity for stages 1–3	$\sim \text{Normal}(0, 1.5^2)$
ω_p	Slope of basal productivity across marine stages	$\sim \text{Normal}(0, 0.5^2)$
$\varepsilon_{p,c}$	Environmental covariate effect	$\sim \text{Normal}(0, \gamma_p^2)$
γ_p	Standard deviation of prior distribution on covariate effect*	$\sim \text{gamma}(1, 1)$
$\tau_{p,c}$	Covariate indicator variable	$\sim \text{Bernoulli}(v_p)$
v_p	Prior inclusion probability for covariate indicator variable*	$\sim \text{Beta}(2, 8)$
$k_{s \in (1:2,3:7),p}$	Basal capacity in log space	$\sim \ln(\text{Uniform}(10^3, 10^7))$
λ_p	Maturation parameter	$\sim \text{Uniform}(0, 5)$
μ_p	Log-normal mean for bycatch mortality rate distribution*	$\sim \text{Uniform}(0, 4)$
v_p	Log-normal standard deviation for bycatch mortality rate distribution*	$\sim \text{Unifrom}(10^{-3}, 2)$
$F_{t \in 1983:2014,p}$	Bycatch fishing mortality rate	$\sim \text{lognormal}(\mu_p, v_p)$
$S_{a \in 1:4,p}$	Bycatch selectivity for 1–4 years at sea	$\sim \text{Uniform}(0, 1)$
$R_{t \in 1980:1986,a,p}$	Initial spawning abundance	$\sim \text{Uniform}(1, 10^4)$
$\sigma_{\text{adult},p}$	Log-normal SD for spawning abundance observation error	$\sim \text{Uniform}(0, 5)$
$\sigma_{\text{juvenile},p}$	Log-normal SD for juvenile index observation error	$\sim \text{Uniform}(0, 5)$
$\sigma_{\text{bycatch},p}$	Log-normal SD for bycatch observation error	$\sim \text{Uniform}(0, 5)$

Note: * indicates model hyperparameters. Population-specific capacity parameters $k_{s,p}$ are estimated for the grouped freshwater (1–2) and marine (3–7) stages.

the proportion of 2 and 3-ocean individuals in the age composition of bycatch.

3.1 | Estimated covariate effects

Covariate effects describe the direction and magnitude of influence each environmental factor is predicted to have on Chinook salmon survival (Figure 3a,b). Generally, there was greater uncertainty in the estimated effect of covariates in the upstream relative to downstream/nearshore and marine stages. While the direction of covariate effects was similar for the two populations, the magnitude of these estimated effects sometimes differed. Strong (95% CI not overlapping zero) negative effects of ice-out date on the Yukon River in the year of ocean entry and of Japanese hatchery chum abundance and strong positive effects of wintertime temperatures at the Pribilof Islands during the first winter at sea are estimated for both populations (Figure 3a,b). During the first year at sea, increased duration of winds, which is favorable for juvenile Pollock growth on the Bering Sea shelf (Megrey & Hinckley, 2001), is also estimated to have a strong positive effect on survival of Chena Chinook salmon, while a higher North Pacific Gyre Oscillation index is associated with increased survival of the Salcha population (Figure 3b).

The effect of wind stress along the Alaska Peninsula in both spring and winter was estimated to be negative for both populations, indicating that marine survival is higher during periods of stronger easterly winds along the peninsula (negative wind stress values), which support a greater flux of nutrient-rich waters northward through Unimak Pass and into the Bering Sea (Stabeno et al., 2002). The two populations differed in the sign of covariate effects for the date of ice out on the natal tributary (Tanana River) in the year of

emergence, for wintertime SLP, and for the Arctic Oscillation index; however, in all three cases the effects were not strong, with 50% or 95% CIs overlapping zero (Figure 3a,b). While the direction and magnitude of covariate effects are informative, they must be considered in conjunction with covariate inclusion probabilities (below).

3.2 | Model selection: covariate inclusion probabilities

The estimated inclusion probability for each environmental covariate (Figure 3c,d) describes the weight of evidence from the data for each hypothesis and can be loosely interpreted as ‘significance’ for a given covariate. For the Salcha population, high (>0.75) inclusion probabilities are estimated for 3 of 21 candidate covariates: Yukon River ice-out date, Pribilof Island wintertime temperature, and Japanese hatchery chum abundance. A larger proportion of covariates exhibited strong support from the data (five of 21) for the Chena population, including the three selected for the Salcha population, plus the Arctic Dipole index during the winter prior to ocean entry and the strength of favorable winds onto the Bering Shelf during the first winter at sea (Figure 3c,d). The prior probability of covariate inclusion (v_p) was updated from its prior expectation (0.2) by each population's data likelihood, with estimated values of 0.38 (95% CI: 0.17–0.63) for the Chena River and 0.35 (0.15–0.60) for the Salcha River.

3.3 | Sensitivity analysis: covariate effects on survival

Covariate effects were quantified by simulating population dynamics while iteratively adjusting covariate values. Figure 4 illustrates both

the percent difference in egg-to-adult survival (a, b) and expected recruits-per-spawner (c, d) when each covariate is increased by one standard deviation from its long-term average while fixing all other covariates at their respective averages. For each 6.6-days later Yukon River ice breakup occurs (1 SD from long-term mean), survival is predicted to be 31.0% lower on average for Chena Chinook salmon, and 28.6% lower for the Salcha population. Chena Chinook salmon survival at ocean entry is also found to be closely associated with variation in the Arctic Dipole wind pattern, with each one standard deviation increase in the index predicted to result in a 27.3% increase in egg-to-adult survival. For wintertime temperature at the Pribilof Islands, each 0.29°C increase is predicted to raise survival of Chena Chinook salmon by 32.4% and Salcha Chinook salmon by 22.6% on average. Competition with Japanese hatchery chum salmon is predicted to have a large impact on survival of both Chinook salmon populations. For each 14.6 million increase in chum adult returns, egg-to-adult survival is expected to be 30.2% lower on average for Chena and 18.9% lower for Salcha Chinook salmon (Figure 4a,b).

3.4 | Habitat capacity, age-specific marine survival, and bycatch selectivity

In addition to environmental effects, other model parameters provide insights into Yukon River Chinook salmon habitat capacity, patterns in marine survival and bycatch mortality. Capacity parameters describe density-dependent limitations on Chinook salmon survival due to intraspecific competition, and were estimated for the upstream and downstream/nearshore (first and second) stages separately from the marine stages. While no capacity limitation is discernable for the marine stages, well-defined capacities for the early life history stages (1–2) are estimated for both populations (Figure 5d,e). In addition to capacity limitations, the population dynamics model also allowed for changes in marine survival as Chinook salmon grow in the ocean through the slope parameter ω_p . Estimates of ω_p are positive for both populations, +0.59 with 95% CI of 0.16–1.31 for the Chena River and +0.50 (0.16–1.04) for the Salcha River, indicating strong evidence for increasing marine survival of Chinook salmon with ocean age (Figure 5c).

Additional marine mortality resulting from BSAI Pollock fishery bycatch was estimated by fitting to bycatch age compositions and Chinook salmon bycatch numbers scaled to the population level. Selectivity of bycatch mortality is estimated to increase with ocean age in both populations, with extremely low bycatch potential for 1-ocean individuals (Figure 5b). In practice, the selectivity of 5-ocean individuals was fixed to be equal to that of four-ocean individuals to improve convergence. Age-specific differences in capture probability may be due to the size-specific selectivity of the gear itself, or more plausibly due to differences in the degree of spatial overlap between salmon and the trawl fleet, with 1-ocean individuals distributed further north and thus at lower risk.

4 | DISCUSSION

Understanding how populations will respond to future environmental change is necessary for risk assessment and conservation planning, yet is contingent upon disentangling the effect that co-occurring natural and anthropogenic factors have on survival. Leveraging all available life history information in a framework that explicitly incorporates multiple sources of uncertainty, our analysis reveals a prominent role of physical and biological drivers in the marine ecosystem. These include interspecific competition with foreign-reared salmon, a very limited effect of bycatch mortality, and evidence that Yukon River Chinook may be subject to density-dependent capacity limitations from intraspecific competition in freshwater but not marine portions of the life cycle. Despite including a wide range of candidate covariates throughout the Chinook salmon lifecycle, we found support for only a small subset of these hypotheses, but this subset was able to explain much of the variation in observed abundance and age composition for these populations (Figure 2).

The only freshwater covariate with strong support from the data was the date of ice breakup on the Yukon River in the year of outmigration, with earlier ice break-up associated with higher juvenile survival. This result is consistent with findings by Ohlberger et al. (2016) and suggests that increased warming of the Yukon River earlier in the season and corresponding earlier onset of primary and secondary production have positive survival benefits for juvenile Chinook salmon. The ecological mechanisms and agents of selection underpinning the association between earlier break-up and survival are unknown, but presumably earlier spring break up results in reduced starvation mortality in early spring, enhanced growth opportunity affording increased survival at ocean entry, beneficial shifts in outmigration timing, or a combination of the above. Although data on how migration timing varies among years is limited for this system, there is mounting evidence that variation in the timing of Chinook salmon ocean entry can have large implications for early marine survival (Satterthwaite et al., 2014; Scheuerell, Zabel, & Sandford, 2009).

The positive association between Chinook salmon survival during the first winter at sea and SST at the Pribilof Islands (Figure 4) suggests that warmer temperatures foster enhanced productivity of these two Chinook salmon stocks. Siegel et al. (2017) provide a potential mechanistic explanation for this relationship, finding a positive correlation between marine temperature and growth of other Chinook salmon populations in this region and suggesting this may result in reduced risk of size-selective marine mortality (Farley, Murphy, Moss, Feldmann, & Eisner, 2009; Murphy et al., 2013). However, our results indicating a positive association between marine survival and temperature for these Chinook salmon populations in the northern portion of Alaska, contrast with findings by Sharma and Liemann (2010) of a net negative association between SST and recruitment of west coast Chinook salmon between Southeast Alaska and Oregon. The idea that salmonid stocks in northern regions respond differently to climate forcing when compared to their conspecifics at southerly latitudes is well established (Hare,

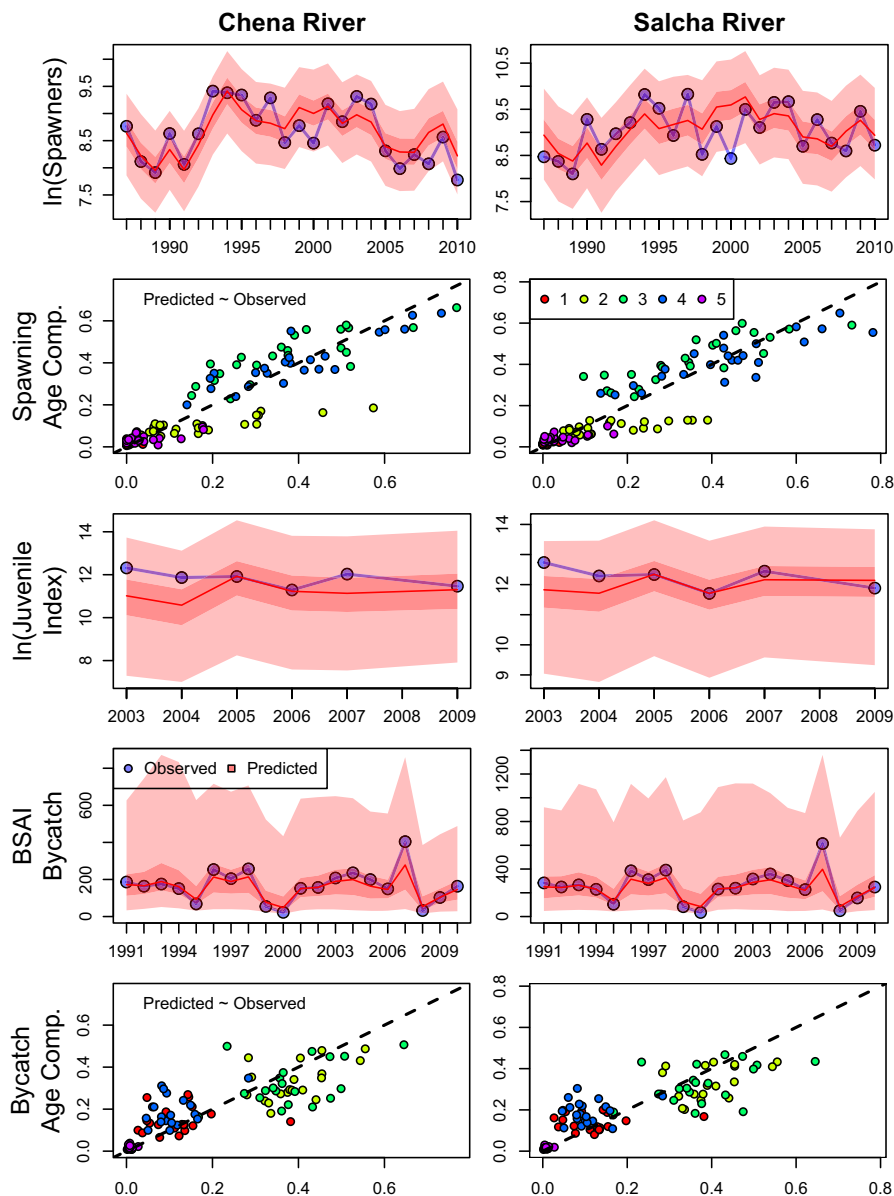


FIGURE 2 Comparison of model predicted adult (spawner) and juvenile abundance indices, bycatch numbers, and age composition, with observed data. Adult and juvenile abundance indices and bycatch data are presented as blue points, with red lines and shaded regions describing the median, 50%, and 95% credible intervals for the posterior predictive distribution. Model fits to age composition data are presented as a comparison of observed and predicted proportions by year and ocean age (year in the ocean). For age composition data, observed proportions are on the x-axis, predicted proportions are on the y-axis, and each point represents a year-by-age class combination. Points falling on the dashed 1:1 line indicate a predicted and observed proportions are equal

Mantua, & Francis, 1999) however, with cooler SST correlated with greater upwelling in the south (Wells, Grimes, Field, & Reiss, 2006; Wells, Grimes, & Waldvogel, 2007). Regardless of the underlying mechanism(s), support for a positive wintertime SST effect implies that a warmer Bering Sea under some climate change scenarios (IPCC, 2014) might enhance future Yukon River Chinook salmon marine survival.

These positive associations between survival and both earlier ice break-up in freshwater and warmer marine temperature conditions during the first winter at sea together suggest that the projected climate warming for this region (Chapin et al., 2014; IPCC, 2014) may

enhance survival of Yukon River Chinook salmon at specific points in the lifecycle, at least for modest increases in the short term. More broadly, these results highlight that Chinook salmon populations at the northern extent of the species' range are expected to respond differently to future climate warming relative to populations in the Pacific Northwest (Battin et al., 2007) and California (Thompson et al., 2012), where responses to a warming climate are predicted to be negative. This aligns with the sentiments of Mantua, Crozier, Reed, Schindler, and Waples (2015) that, "the impacts of future climate warming on Pacific salmon will be mixed, and context dependent". While our results suggest increased survival with local and

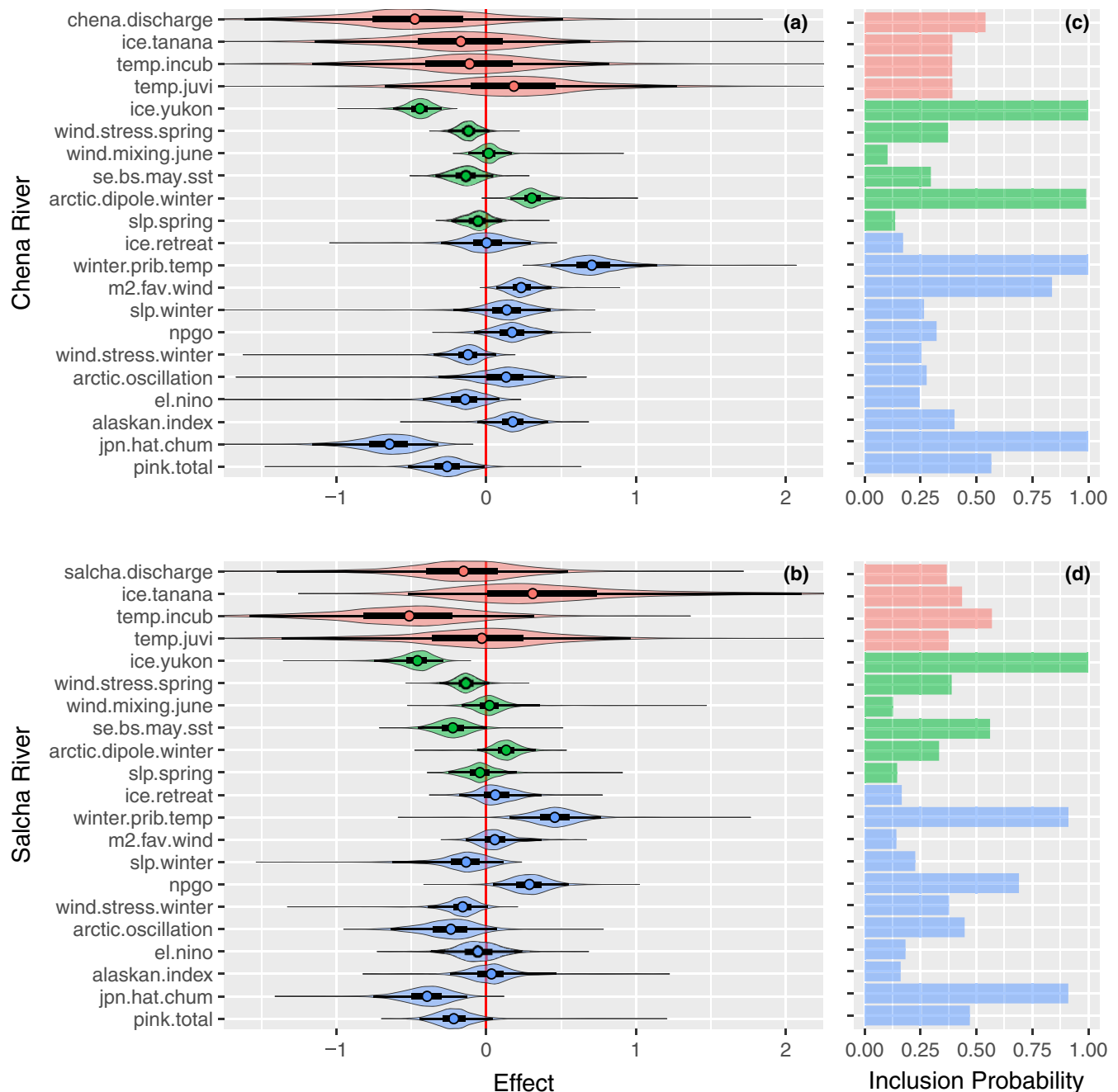


FIGURE 3 Covariate effects and inclusion probabilities. Left panels (a, b) show posterior distributions for covariate effects as caterpillar plots where the point, thick, and thin lines, indicate the median, 50% and 95% credible intervals. Posterior distributions for covariate effects are only samples for which the covariate was included ($\tau_{p,c} = 1$). Right panels (c, d) show inclusion probability for each covariate. For descriptions of covariates, see Table 1. Colors correspond to model stage, red being upstream, green is downstream/nearshore marine, and blue is offshore marine

regional scale climate warming, it is critical to remember that: (a) these relationships only represent the range of data observed historically and may not fully capture potential nonlinear responses to temperature regimes outside the range of past observations, (b) temperature effects at other points in the lifecycle may have different effects on survival, (c) future climate change will alter aspects of marine and freshwater environments other than temperature, and changes in other habitat variables may have negative impacts, (d)

climate-mediated northward shifts in the distribution of marine species (Pinsky, Worm, Fogarty, Sarmiento, & Levin, 2013; Thorson, Pinsky, & Ward, 2016) may result in increased predation risk to juvenile and adult Chinook salmon, and (e) temperature may only be a proxy for the actual driver of survival variation.

In addition to climate effects, the model also detected a negative association between the Japanese hatchery chum abundance and marine survival of Yukon River Chinook salmon (Figure 4a,b). Although

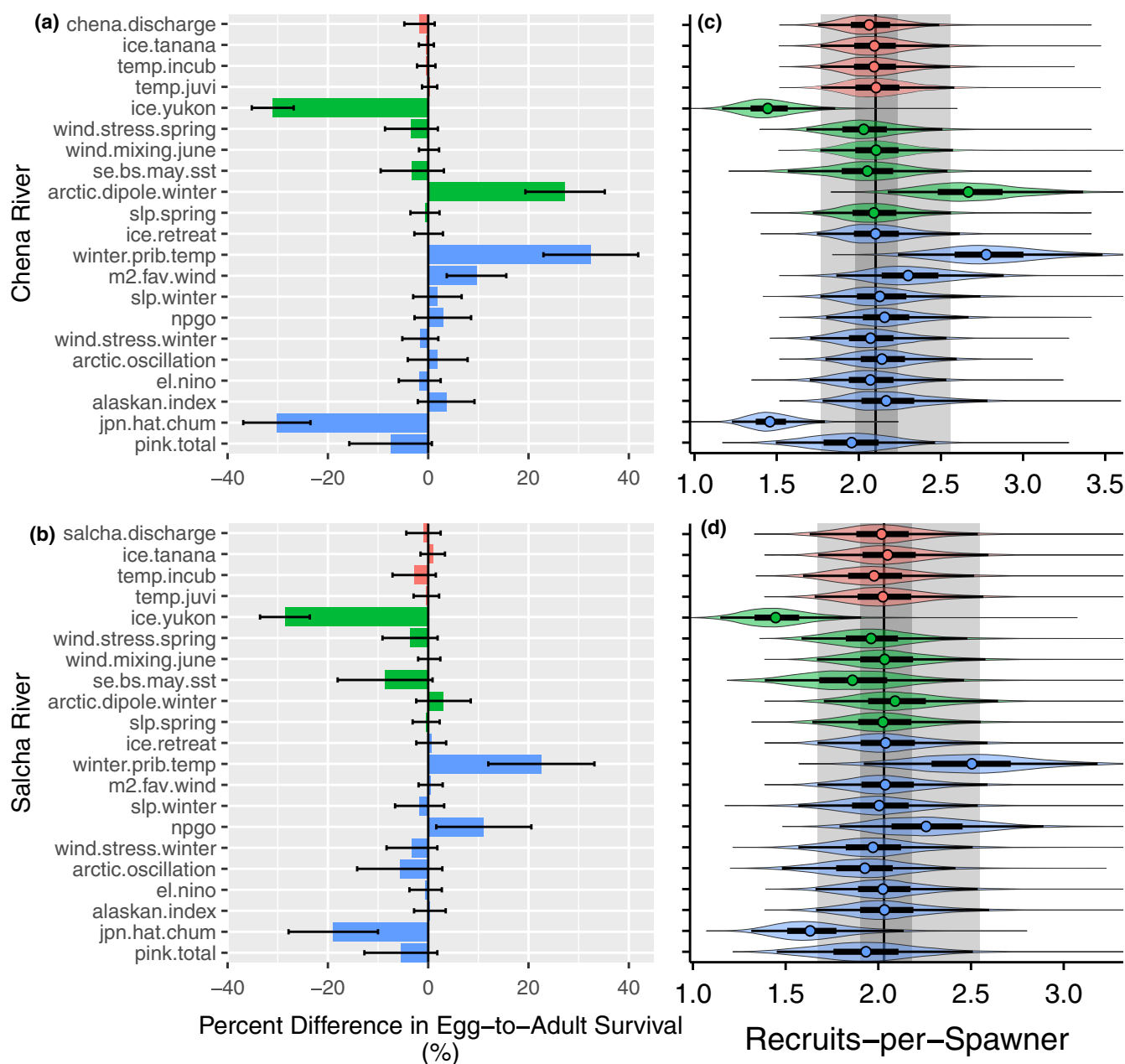


FIGURE 4 Simulated covariate effects. Left panels (a, b) show the expected percent change in egg-to-adult survival when each covariate is increased by +1 standard deviation from its long-term average. Right panels (c, d) show the distribution of expected productivity (recruits-per-spawner) for each simulated case, relative to the base case (all covariates are fixed at their long-term average). The black line, dark, and light gray shaded regions, denote the median, 50% and 95% credible intervals for base case productivity. Colors correspond to model stage (Table 1, Figure 1)

the effect of increased pink salmon abundance was also estimated to be negative for both populations, there was weak support for this hypothesis (Figure 3). The observation of decreasing survival with increasing hatchery production is generally consistent with possible density-dependent effects on survival at different spatial scales during the marine phase for various Pacific salmon species (Peterman, 1982; Ruggerone, Peterman, Dorner, & Myers, 2010). Ruggerone and Nielsen (2005) found that increased pink salmon abundance reduced survival of Puget Sound Chinook salmon and growth of Bristol Bay sockeye salmon through competition for prey resources, while

Ruggerone and Connors (2015) found North Pacific pink salmon abundance to be negatively correlated with sockeye survival and size-at-age and positively correlated with later age at maturity. These results suggest that increased hatchery production of chum salmon, whether by Japan or other countries, could reduce Yukon River Chinook salmon survival.

Although with less support from the data, the estimated direction of effects for other environmental covariates is also informative. For example, the effect of wind stress on Chinook salmon survival at ocean entry and during the first winter at sea was negative for both

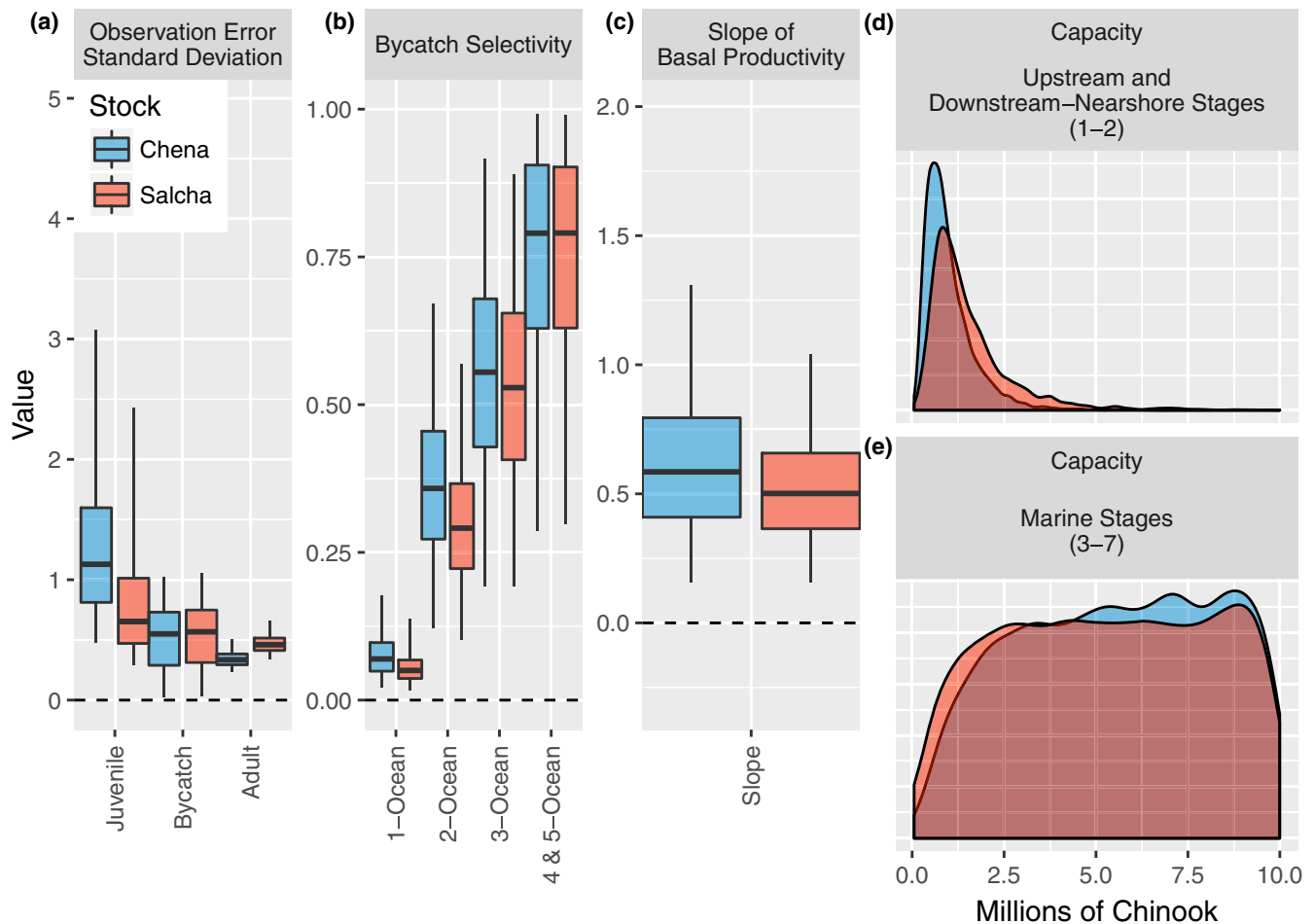


FIGURE 5 Posterior distributions for other population dynamics model parameters: (a) standard deviation of the lognormal observation error for the three indices of abundance σ_{p} , (b) marine bycatch selectivity by ocean age $S_{a,p}$, (c) slope of the change in basal productivity (maximum survival rate) across marine stages ω_p , and (d, e) capacities for upstream and nearshore stages and marine stages $k_{s,p}$

populations (Figure 3a,b). Negative wind stress values denote anomalously strong easterly winds that drive northward transport of nutrient-rich waters through Unimak Pass (Stabeno et al., 2002). Wind-driven northward transport of nutrients, particularly nitrogen, plays an important role in modulating productivity of the Bering Sea food web (Rosenkranz, Tyler, & Kruse, 2001; Stabeno et al., 2002). These negative effects of wind stress on Chinook salmon survival concur with our a priori hypothesis, despite the relatively low inclusion probabilities and limited utility in forecasting future survival. Consistent with Neuswanger et al. (2015), our results suggest increased water discharge in natal streams during the summer following emergence is associated with lower juvenile survival (Figure 3a,b), although discharge also had a low inclusion probability (Figure 3c,d). Neuswanger et al. (2015) hypothesized this negative association with arise from reduced foraging efficiency of juvenile Chinook as a result of increased water turbidity during high flows, though the exact mechanism(s) remain unclear (Neuswanger, Wipfli, Rosenberger, & Hughes, 2014).

Estimated effects of upstream (first stage) covariates had both high uncertainty and low inclusion probabilities (Figure 3), together indicating little evidence that survival to the adult stage has responded in a predictable manner to freshwater environmental

conditions early in the life cycle (Figure 4). However, it is important to note that this finding might result from several processes. First, freshwater processes might indeed not drive variation in survival. Second, downstream survival bottlenecks might mask the influence of upstream processes and therefore our ability to discern their effect. Third, the lack of abundance indices for juveniles exiting tributaries and entering the mainstem Yukon (between stages 1 and 2) limits our ability to partition mortality between upstream and downstream/nearshore environments. Finally, we may have failed to include the most relevant freshwater covariates due to the limited monitoring in these large, remote watersheds. In any case, results suggest that future trends in survival are better predicted by marine rather than freshwater conditions, but additional abundance indices for juvenile Chinook as they exit tributaries or for Chinook smolt as they enter the ocean may be of value in improving understanding of population dynamics.

In conclusion, the use of stage-structured models in conjunction with Bayesian model selection methods provides a useful framework for testing hypotheses about, and quantifying the influence of, environmental processes on survival of species at different locations in space and time, throughout their life cycle, and generating

predictions that are robust to uncertainty in both estimation and model structure. However, these methods are data intensive and are correlative in nature and thus limited by contrast in both covariate and biological data. Applications of these methods to Yukon River Chinook salmon populations suggest that with the exception of variables related to the timing of springtime ice breakup in freshwater, processes in the marine environment better explain survival variation. This does not suggest, however, that freshwater processes are unimportant as these populations' genomes likely reflect site-specific adaptations to selection pressures of freshwater habitats that bolster population robustness and stability (Carlson & Satterthwaite, 2011; Griffiths, Schindler, & Seeb, 2013; Hilborn, Quinn, Schindler, & Rogers, 2003). The maintenance of a rich portfolio of complex, diverse, and interconnected freshwater habitats in the middle Yukon River and beyond should be a conservation goal (Carlson & Satterthwaite, 2011; Griffiths et al., 2014). Finally, results suggesting density-dependent capacity limitations in freshwater indicate that any proposed enhancement programs designed to benefit depressed Yukon River stocks should carefully consider potential impacts on wild populations and probability of success given downstream bottlenecks, while also considering the global connectivity of Pacific salmon rearing in the North Pacific Ocean.

Perhaps most importantly, from the perspective of future climate change, the strong support for positive association between Yukon River Chinook salmon survival and both earlier ice breakup in freshwater and warmer wintertime temperatures at sea suggests that in the short term future climate warming is more likely to have a net positive impact on these Chinook populations at the northern extent of their Pacific range, in contrast with predicted impacts on populations inhabiting southerly portions of the species' range in the North Pacific (Battin et al., 2007; Crozier et al., 2008; Thompson et al., 2012). However, this prediction is based on the range of environmental conditions observed historically and may not be applicable if and when conditions exceed the range of past observations.

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ORCID

Curry J. Cunningham  <http://orcid.org/0000-0002-1234-1297>

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SUPPORTING INFORMATION

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